

A further conjecture on OEIS A224412, settled by the proof already in hand

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Abstract

OEIS A224412 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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1 The premise

The entry's first terms are 32, 324, 1288, 3436, 7812, 16420, 32814, 63202, $\dots$, with offset 1. The sequence satisfies

$$a(n) = 11a(n-1) - 55a(n-2) + 165a(n-3) - 330a(n-4) + 462a(n-5) - 462a(n-6) + 330a(n-7) - 165a(n-8) + 55a(n-9) - 11a(n-10)$$

a linear recurrence of order 11 with characteristic polynomial

$$q(x) = x^{11} - 11x^{10} + 55x^9 - 165x^8 + 330x^7 - 462x^6 + 462x^5 - 330x^4 + 165x^3 - 55x^2 + 11x - 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least $11 + 2$ further terms after that index.

2 The further conjecture

Separately from the conjecture already settled, the entry records

$$\begin{aligned} \text{Empirical: } a(n) = & (1/3628800)*n^{10} + (1/80640)*n^9 + (1/3456)*n^8 + \\ & (23/5760)*n^7 + (7213/172800)*n^6 + (683/2304)*n^5 + (138869/36288)*n^4 + \\ & (32143/2240)*n^3 + (32909/450)*n^2 - (753/20)*n - 44 \text{ for } n \end{aligned}$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that a and the closed form f satisfy the same linear recurrence; it does not say they are the same solution of it. Write p for the annihilator of f ,

$$p(x) = p(x),$$

of order $D = 11$. Since $q \mid p$ and a satisfies the recurrence of q , a satisfies that of p as well; f satisfies it by construction. Two solutions of a linear recurrence of order D with nonzero leading coefficient agree everywhere as soon as they agree at D consecutive indices, and a and f were checked to agree at $n = 15, \dots, 25$, which lies inside the terms the entry publishes and past the index from which a provably satisfies p . Hence $a(n) = f(n)$ for every n in the claimed range.

Theorem 1. *The conjecture displayed above is true.*

3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly q says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.